

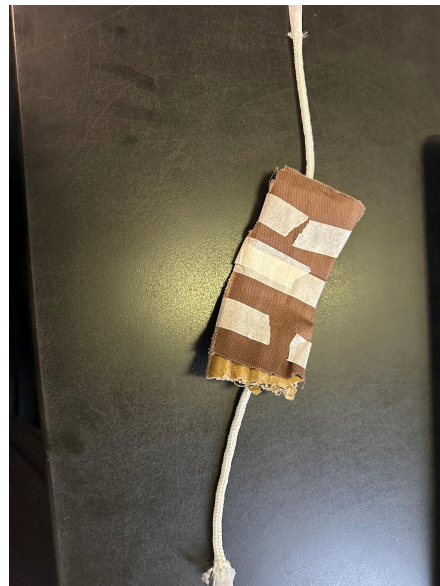
A Self-Closing Gate Conundrum

#1 Victory Royale

Element G: Construction of a Testable Prototype

Student G's family needs a self-closing mechanism that closes their family's gate on its own because his family does not want unwanted animals and people to get inside their fence gate because it is a safety concern that could cause injury. This happens because student G's gate can not close on its own.

We cut out a piece of cardboard to make it fit the size of student G's fence and then attached it using string tape on the back to be able to tie it around student G's fence and tighten it. We put a pad of cloth on the cardboard and attached it with tape. The changes we made was we needed padding in between the fence and the clasp so the metal clasp does not scratch the paint off the fence.



We started by cutting our galvanized hanger strap to the correct length and then putting a

bolt in it to tighten it to the perfect size. We attached the tension spring and the clamp to the fence and the gate so that it will close on its own. It will help us collect data on our design requirements by showing us what will be able to be done and see what needs to change to make us meet all the requirements.



For gate closing power we know that the spring is strong enough to pull the gate close all the way. For testing weather resistance we bought only weather resistant products so they will be weather resistant. We can not test durability because the prototype we made was made out of cardboard which is completely different from metal. We added up the total cost of our items and it is in the budget. There are a lot of parts that can not be tested with our first prototype like the gate closing because we do not have our spring. Our final prototype we can do everything because we have all of our items like the spring and the clasp.